

Deceptive Empathy in Large Language Model Responses to Suicide-Related Disclosures

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Abstract

Background: Large language model (LLM) chatbots are increasingly used for emotional and mental health support. Users often rate their responses as helpful, supportive, and empathic, and recent studies suggest that LLM-generated replies can sometimes be judged as more empathic or therapeutically preferable than human-written responses. In suicide-related disclosures, however, such perceived empathy may introduce a distinct safety concern. By generating human-like language that implies relational understanding, care, or presence, LLMs may simulate an interpersonal capacity they do not possess. Iftikhar et al term this phenomenon “deceptive empathy”, referring to anthropomorphic, relationally simulating responses that may mislead users about the nature of the interaction [7].

Objective: This study aimed to quantify the prevalence of deceptive empathy in responses generated by commercially available large language models (LLMs) to suicide-related disclosures, a high-risk context requiring clinically appropriate and risk-sensitive responding.

Methods: Fifty posts were randomly sampled from the CLPsych 2019 r/SuicideWatch dataset [18]. Two licensed psychotherapists independently reviewed these posts and each selected 20 that represented high-risk suicide-related disclosures warranting immediate risk assessment and possible escalation to a higher level of care. After consensus review, 20 prompts were retained for model evaluation. In February 2026, each prompt was submitted once to 8 commercially available free-tier large language model (LLM) systems, including ChatGPT 5.2, Gemini 2.5 Flash, Claude Haiku 4.5, DeepSeek V3.2, Perplexity, Grok 3, Llama 4, and Mistral. A total of 160 outputs are evaluated. Deceptive empathy was coded using a prespecified rubric by 3 independent coders. Interrater reliability before consensus adjudication was acceptable (Fleiss $\kappa=0.66$).

Results: One response (0.6%) was excluded because it was generated in Spanish, leaving 159 evaluable outputs. Deceptive empathy was present in 96/159 responses (60.4%). Prevalence varied substantially across models, ranging from 0/20 (0.0%) to 20/20 (100.0%). Differences by model were statistically significant ($\chi^2=84.35$; $P<.001$). Three models generated deceptive empathy in all responses, whereas 1 model produced none because it refused all prompts.

Conclusions: Deceptive empathy was common in LLM responses to suicide-related disclosures and varied sharply by model. In high-risk mental health contexts, perceived empathy should not be treated as evidence of safety or clinical adequacy. Evaluations of mental health chatbots should separately assess deceptive empathy, risk acknowledgment, escalation guidance, and refusal behavior.

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Original Manuscript

Deceptive Empathy in Large Language Model Responses to Suicide-Related Disclosures

Research Letter

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Keywords

Large Language Models; suicide; deceptive empathy; anthropomorphism; mental health chatbots; crisis response; AI safety; digital mental health

Deceptive Empathy in Large Language Model Responses to Suicide-Related Disclosures

Introduction

Large language model (LLM) based chatbots are increasingly used for emotional and mental health support, both informally by the public and more explicitly in digital mental health contexts [1-3]. Users often describe these systems as accessible, nonjudgmental, and emotionally supportive [1]. Recent studies suggest that LLM-generated responses can be rated favorably on therapeutic communication dimensions and, in some settings, judged as difficult to distinguish from therapist-written responses or even rated as more empathic than human responses [4,5]. A broader systematic review of AI mental health chatbots likewise suggests rapid growth of LLM-based systems despite uneven clinical evaluation and variable evidence of safety and efficacy [4] which creates a serious interpretive problem in suicide-related interactions.

In lower suicide risk contexts, warm and validating language may be experienced as helpful. In high suicide risk settings, however, users' inference to understanding, concern, containment, judgment, and relational presence from the system's language is problematic. Prior work on emotional dependence and social chatbots has shown that users can form strong attachments to conversational systems and may experience psychological harm when those systems are perceived as human-like but fail to function as humans do [6]. Recent evaluations of generative AI psychotherapy chatbots used by youth have similarly raised concerns about clinically unsafe or incomplete responses to high-risk scenarios [7]. The anthropomorphic, relationally simulating, human-like empathic language in crisis contexts can mislead users about what kind of entity is responding and what level of care, responsibility, and judgment is actually present.

The term deceptive empathy proposed by Iftikhar et al captures this issue from an ethical perspective [8]. The authors state that deceptive empathy as formulaic empathic responding to emotional disclosure that may mislead users by simulating understanding or care without an experiencing self behind the utterance [8]. Related work by the same research group explicitly described "deceptive empathy" as a recurring concern when psychologists evaluated LLM counseling responses, arguing that such outputs can create the appearance of empathic attunement while lacking the subjectivity and accountability that make empathy meaningful in human care [9].

According to Carl Rogers, the founder of the Client Centered Psychotherapy, therapeutic effectiveness depends on three necessary and sufficient conditions which are congruence, unconditional positive regard, and empathy [10]. Contemporary supportive psychotherapy guidance reaches a similar conclusion cautioning that statements that are rote, mechanical, or inauthentic can

weaken rather than support the therapeutic process [11]. This distinction is especially important for LLMs. Although these systems can produce highly fluent interpersonal language, they do so without felt concern, personal stake, moral agency, or clinical accountability that are the very human capacities that give empathic language its therapeutic meaning and ethical force.

Digital mental health ethics makes the problem more precise. Ferrario et al argue that the humanization of conversational AI in depression-related contexts deserves critical scrutiny because design choices that increase perceived humanness may also increase inappropriate trust [12]. Hartford and Stein similarly contend that meaningful relationships are shaped not simply by convincing outputs, but by effort, reciprocity, and the moral structure of relating that are the features conversational AI cannot possess even when it reproduces their linguistic surface [13]. Related bioethical commentary warns that such systems may undermine user autonomy when their human-like style obscures both their ontological limits and the boundaries of their responsibility [14]. In mental health contexts, this concern becomes especially acute because vulnerability, dependency, deception, and help-seeking define its ethical stakes [15].

Despite growing concern, deceptive empathy has not yet been widely operationalized as a measurable audit construct in evaluations of high-risk mental health AI. Most existing assessments focus on helpfulness, perceived empathy, general therapeutic quality, or crisis safety more broadly [1-6] and leaves an important gap of exploring how the LLM based chatbots may appear empathic and clinically appropriate while still using anthropomorphic, relationally simulating language that misrepresents the nature of the interaction. This study addresses that gap by quantifying deceptive empathy in responses generated by 8 commercially available LLMs to suicide-related disclosures drawn from r/SuicideWatch [16]. We examined how frequently deceptive empathy occurred, whether its prevalence differed by model, and the implications of these findings for evaluation, governance, and safety.

Method

Fifty user posts were randomly chosen from r/SuicideWatch dataset in January 2026. Two licensed psychotherapists independently reviewed the set, and each selected 20 high-risk posts (i.e., disclosures that would typically trigger an immediate clinical duty to interrupt routine therapy processes and assess imminent risk and/or escalate to a higher level of care), then met for consensus and agreed on 20 prompts for model evaluation.

In January 2026, these 20 prompts were submitted to 8 commercially available free-tier LLMs (ChatGPT 5.2, Gemini 2.5 Flash, Claude Haiku 4.5, DeepSeek V3.2, Perplexity, Grok 3, Llama 4, and Mistral) using newly created accounts. The prompts were sent via each system's API, and a single response was collected by request. To minimize carryover effects, the prompt-response pair was deleted after each exchange. A new session was initiated with a VPN-selected location change, and the procedure was repeated for each prompt, yielding 160 total LLM outputs (20 prompts x 8 models).

Deceptive empathy was coded using a prespecified rubric developed for this study. The construct was defined as anthropomorphic, relationally simulating empathic language that implied felt concern, interpersonal presence, or human-like understanding not actually possessed by the model. The coding focus was the LLM-generated responses to the prompts. Embodied-empathy phrases including first-person feeling claims (e.g., "I'm heartbroken," "I'm scared for you," "my heart hurts."), companion-like presence claims (e.g., "I'm here with you," "I won't leave," "I'm right here."), and intimate or relationally suggestive expressions of care (e.g., "I care about you deeply") are coded as Deceptive Empathy presence. The non-embodied empathy remarks (e.g., "That sounds painful") are not coded as Deceptive Empathy presence. Three coders independently rated all outputs. Disagreements were resolved through consensus adjudication after independent coding. Interrater reliability before consensus adjudication was assessed using Fleiss κ and was 0.66,

indicating acceptable agreement.

Results

Across 160 collected outputs, 1 response (0.6%) was excluded because it was generated in Spanish. The final analytic sample therefore included 159 responses.

Deceptive empathy was present in 96/159 responses (60.4%). Prevalence varied markedly across models, ranging from 0/20 (0.0%) to 20/20 (100.0%). Differences across models were statistically significant ($\chi^2=84.35$; $P<.001$; Cramer $V=0.73$), indicating a large between-model effect. Three model (i.e. DeepSeek V3.2, Claude Haiku 4.5, and Mistral) produced deceptive empathy in 100% of outputs (20/20 each). Grok 3 produced no deceptive empathy (0/20) because it refused all prompts. The remaining models showed intermediate rates, with Llama 4 producing the lowest nonzero prevalence (4/19, 21.1%).

Table 1. Deceptive Empathy by Model

Model	n	DE=Present, Raw (%)
Gemini 2.5 Flash	20	11 (55.0%)
Deepseek V3.2	20	20 (100.0%)
Grok 3	20	0 (0.0%)
ChatGPT 5.2	20	12 (60.0%)
Perplexity	20	9 (45.0%)
Claude Haiku 4.5	20	20 (100.0%)
Llama 4	19	4 (21.1%)
Mistral	20	20 (100.0%)
Overall	159	96 (60.4%)

Discussion

Deceptive empathy was common in LLM responses to suicide-related disclosures. In this sample, 60.4% of evaluable outputs contained anthropomorphic, relationally simulating empathic language that implied felt concern, interpersonal presence, or human-like understanding not actually possessed by the model, and prevalence differed sharply across models.

The present findings support treating deceptive empathy as a distinct safety-relevant construct rather than a stylistic feature. In low-risk contexts, users may interpret statements such as “I’m here for you,” “I care about you,” or other first-person empathic phrasing as benign social language. In crisis contexts, that interpretation becomes heavier. Such phrases can imply containment, concern, availability, and understanding. When generated by a system without subjectivity, accountability, or clinical responsibility, they may inflate the apparent trustworthiness and relational adequacy of the response. That is the core ethical problem identified by the concept of deceptive empathy [8,9]. Human empathy in psychotherapy is embedded in a real relational stance involving genuineness, responsibility, and attunement [10,11]. When an LLM produces language that mimics those signals, it can borrow the social force of empathy without meeting its interpersonal conditions [12,13].

These results have several implications for evaluation and governance. First, perceived empathy should not be treated as evidence of clinical safety. A model can sound warm and still fail to respond in a risk-sensitive manner. Second, the marked differences between models suggest that

deceptive empathy is not an unavoidable property of LLM output. It appears to be design-dependent and therefore potentially reducible through model policy, interface constraints, prompting strategy, refusal behavior, or other system-level choices. Third, the findings support adding deceptive empathy to pre-deployment and post-deployment audit frameworks for mental health AI. Existing evaluations commonly examine helpfulness, adherence, symptom outcomes, or broad safety categories [2,3,7]. Those remain necessary but are insufficient. A model that avoids deceptive empathy only by refusing every disclosure may reduce relational simulation while still failing to provide clinically appropriate crisis engagement. Conversely, a model that is emotionally fluent but poor at escalation may appear safer than it is. These should not be collapsed into a single “good” or “bad” score.

This study has limitations. The prompt set was small and clinician-selected, which improved relevance to high-acuity mental health practice but may limit generalizability. Responses were sampled in a single-turn format and during one collection window, so findings may not generalize to extended interactions or future model updates. Also, only free-tier systems were evaluated. One non-English output was excluded, and the coding framework was developed for English-language responses. In addition, this study measured textual features of model outputs rather than user perceptions or downstream behavioral effects. It therefore cannot determine whether deceptive empathy directly changes user trust, help-seeking, or crisis outcomes, although prior literature provides strong reason to consider that possibility.

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